# JOSE MANUEL BARCENA ANIDO


**Software Engineer - Python, C/C++, Cloud & AI Systems**

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• Languages: Spanish (native) · Galician (native) · English (C1) · French (B1)


## PROFILE

Software Engineer combining enterprise cloud/data experience with strong Computer Science depth across systems, compilers, backend engineering and AI/ML. Delivered production migration and validation work across Azure, Databricks, Snowflake and SQL; built a C/C++ formal-language compiler adopted in university research collaboration involving NASA Langley; and developed LLM automation, inference and physiological-signal ML systems. Currently tutors programming and Computer Science from vocational through MSc level.

## CORE ENGINEERING STACK

### Programming & Backend
Python, C++, C#, C, FastAPI, REST APIs, async workflows, OpenAI-compatible APIs, OOP

### Cloud & Data
Azure, Databricks, Snowflake, SQL, schema design, data modelling, ETL pipelines, Docker, Linux, CI/CD

### AI & ML Systems
PyTorch, Llama 3.1, Ollama, vLLM, model serving, CNNs, physiological-signal ML, inference benchmarking

### Engineering Quality
Automated/integration testing, semantic/type validation, error recovery, GDB, Valgrind, CMake, Git, Jira

## SELECTED ENGINEERING PROJECTS

**Alfred - Local AI Automation System** | Python, Ollama, Llama 3.1
github.com/Josebarcena/Alfred

Engineered a modular local AI automation system that converts natural-language requests into structured, validated tool executions with deterministic routing, argument validation, isolated execution, timeouts and inspectable responses.

**LLM Inference Performance Lab** | Python, PyTorch, vLLM, NVIDIA GPU
github.com/Josebarcena/LLM-Inference-Performance-Lab

Built an asynchronous evaluation harness around an OpenAI-compatible vLLM service; cut input padding from 57.8% to 5.3% and improved output throughput by ~12% while tracking TTFT, latency, GPU memory, power and stability.

**TPTP to PVS Compiler & Validation Pipeline** | C/C++, Flex/Bison
github.com/Josebarcena/TPTP-PVS

Designed and implemented a complete source-to-source compiler pipeline with parsing, ASTs, symbol tables, semantic/type validation, error recovery and code generation; achieved 95.6% translation fidelity.

Adopted by faculty researchers in collaboration with the University of Nebraska Omaha and NASA Langley Research Center.

**ECG Anomaly Detection** | Julia, Flux, CNNs, PTB-XL
github.com/Josebarcena/DeepLearning

Built an end-to-end ML pipeline for human physiological ECG data, covering preprocessing, reproducible training/validation and metrics export; achieved >90% F1-score and precision.

## ENGINEERING EXPERIENCE

**Data & Software Engineer Intern | Indra** — **Apr 2025 - Aug 2025**

- Joined through a junior-engineer internship and ramped quickly into production delivery, implementing cross-system validation and backend integration for a leading global fashion retailer across Azure, Databricks, Snowflake and SQL.
- Optimized distributed workloads, reducing execution time and operational costs by approximately 30%, while coordinating defects and validation findings through Jira.

**IT Operations & Systems Technician | Softtek** — **Apr 2019 - Aug 2019**

- Supported enterprise environments for General Electric, Inditex, ABANCA and Electronic Arts across Linux/Windows systems, SQL-backed applications, deployments, accounts and networks; collaborated with operations teams while resolving 150+ incidents.

## ADDITIONAL EXPERIENCE

**Programming & Computer Science Tutor | Independent / Part-time** — **2026 - Present**

- Tutored 10 vocational, BSc and MSc students across programming, algorithms, software engineering, systems and debugging, adapting languages and technologies to individual coursework; 7 achieved top marks in practical programming assignments.

## EDUCATION

**MSc in Software Engineering | UNIR** — **2025 - 2026**
Modern architecture, backend/API design, cloud systems, software quality and testing. Coursework completed; thesis remaining.

**BSc Computer Engineering - Computation | Universidade da Coruña** — **2020 - 2025**
260/240 ECTS completed · Algorithms, AI, formal methods, programming languages, software engineering and computational problem solving.

**Advanced Vocational Training - Systems Administration & Networking** — **2017 - 2019**
Linux/Windows administration, networking, databases, infrastructure, deployment and enterprise troubleshooting.